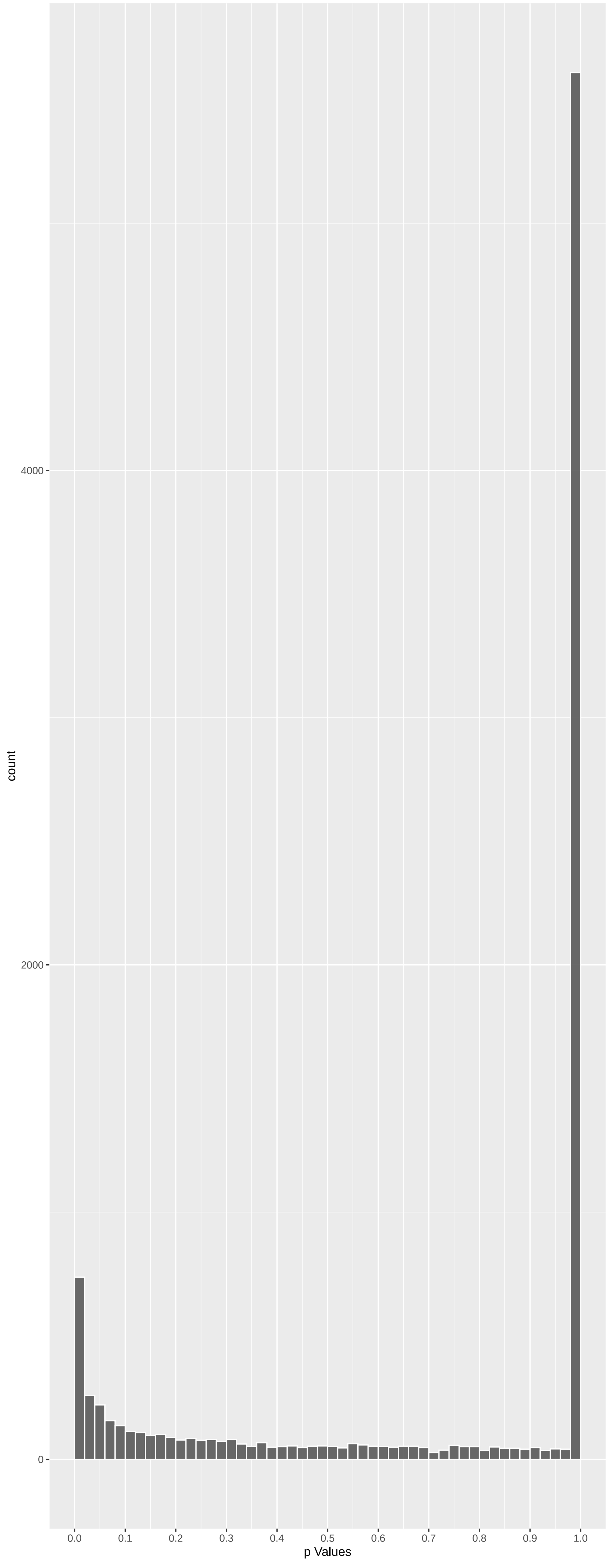
Top genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	−6.491090	5.113050e−10	7.723e−06	1.166e−01
PROM1	5.975173	1.379080e−08	1.041e−04	3.699e−01
HIRA	5.796892	4.053315e−08	1.224e−04	3.699e−01
SPINT4	5.809237	3.765500e−08	1.224e−04	3.699e−01
BNC1	−5.863049	2.726658e−08	1.224e−04	3.699e−01
TRPV1	5.692094	7.529438e−08	1.895e−04	4.771e−01
RPL23	5.516157	2.078961e−07	4.486e−04	6.886e−01
ATAD3C	5.485062	2.479943e−07	4.682e−04	6.886e−01
CLDN16	5.441644	3.167459e−07	5.021e−04	6.886e−01
PPP4C	5.433043	3.324059e−07	5.021e−04	6.886e−01
SNRPD1	−5.345258	5.417301e−07	7.438e−04	6.886e−01
AFP	5.283333	7.611261e−07	8.547e−04	6.886e−01
TUBB8B	5.274319	7.994997e−07	8.547e−04	6.886e−01
TRA2B	5.263323	8.488468e−07	8.547e−04	6.886e−01
TUBB8	5.293369	7.204990e−07	8.547e−04	6.886e−01
CELA3A	5.189181	1.267324e−06	9.658e−04	6.886e−01
SULT1A2	5.213602	1.111251e−06	9.658e−04	6.886e−01
UBE2H	−5.188551	1.271622e−06	9.658e−04	6.886e−01
H3−3A	5.214582	1.105395e−06	9.658e−04	6.886e−01
RPS14	5.187487	1.278905e−06	9.658e−04	6.886e−01
UBE2D1	−5.154457	1.526204e−06	1.098e−03	6.886e−01
ANO5	5.145556	1.600374e−06	1.099e−03	6.886e−01
UBE2B	5.105996	1.974339e−06	1.297e−03	6.886e−01
PGBD1	−5.096869	2.071900e−06	1.304e−03	6.886e−01
SF3B6	−5.016845	3.151613e−06	1.831e−03	6.886e−01

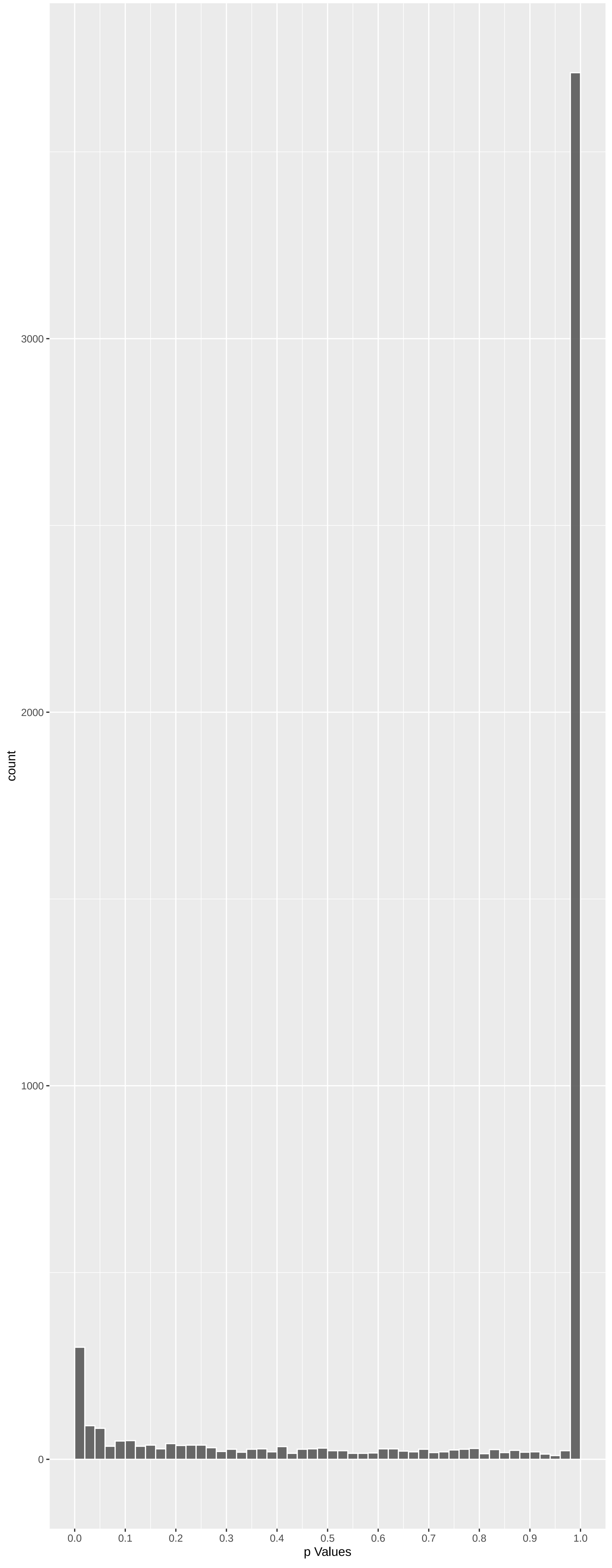
Top genes by Q−Value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	−6.491090	5.113050e−10	7.723e−06	1.166e−01
HIRA	5.796892	4.053315e−08	1.224e−04	3.699e−01
SPINT4	5.809237	3.765500e−08	1.224e−04	3.699e−01
BNC1	−5.863049	2.726658e−08	1.224e−04	3.699e−01
PROM1	5.975173	1.379080e−08	1.041e−04	3.699e−01
TRPV1	5.692094	7.529438e−08	1.895e−04	4.771e−01
CEMIP	−4.964176	4.139594e−06	1.955e−03	6.886e−01
AFP	5.283333	7.611261e−07	8.547e−04	6.886e−01
SF3B6	−5.016845	3.151613e−06	1.831e−03	6.886e−01
TAS2R10	4.691656	1.626016e−05	3.411e−03	6.886e−01
PGLS	4.705640	1.518423e−05	3.354e−03	6.886e−01
BPIFB1	4.985994	3.698642e−06	1.908e−03	6.886e−01
SNAP25	−4.929729	4.940626e−06	1.987e−03	6.886e−01
NARS1	4.963700	4.149757e−06	1.955e−03	6.886e−01
SDS	4.754803	1.191839e−05	2.994e−03	6.886e−01
PGBD1	−5.096869	2.071900e−06	1.304e−03	6.886e−01
UBE2K	4.730274	1.345303e−05	3.209e−03	6.886e−01
CFAP20	−4.839052	7.827600e−06	2.627e−03	6.886e−01
CLDN16	5.441644	3.167459e−07	5.021e−04	6.886e−01
UBE2B	5.105996	1.974339e−06	1.297e−03	6.886e−01
ADAM2	4.990673	3.610160e−06	1.908e−03	6.886e−01
PPP4C	5.433043	3.324059e−07	5.021e−04	6.886e−01
PKD1L1	−4.808859	9.107662e−06	2.711e−03	6.886e−01
CELA3A	5.189181	1.267324e−06	9.658e−04	6.886e−01
ACTC1	−4.691812	1.624779e−05	3.411e−03	6.886e−01

Positive Rho Non−permulated



Negative Rho Non−permulated



Top Positive genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
PROM1	5.975173	1.379080e−08	1.041e−04	3.699e−01
HIRA	5.796892	4.053315e−08	1.224e−04	3.699e−01
SPINT4	5.809237	3.765500e−08	1.224e−04	3.699e−01
TRPV1	5.692094	7.529438e−08	1.895e−04	4.771e−01
RPL23	5.516157	2.078961e−07	4.486e−04	6.886e−01
ATAD3C	5.485062	2.479943e−07	4.682e−04	6.886e−01
CLDN16	5.441644	3.167459e−07	5.021e−04	6.886e−01
PPP4C	5.433043	3.324059e−07	5.021e−04	6.886e−01
AFP	5.283333	7.611261e−07	8.547e−04	6.886e−01
TUBB8B	5.274319	7.994997e−07	8.547e−04	6.886e−01

Top Negative genes by P−value non−permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	−6.491090	5.113050e−10	7.723e−06	1.166e−01
BNC1	−5.863049	2.726658e−08	1.224e−04	3.699e−01
SNRPD1	−5.345258	5.417301e−07	7.438e−04	6.886e−01
UBE2H	−5.188551	1.271622e−06	9.658e−04	6.886e−01
UBE2D1	−5.154457	1.526204e−06	1.098e−03	6.886e−01
PGBD1	−5.096869	2.071900e−06	1.304e−03	6.886e−01
SF3B6	−5.016845	3.151613e−06	1.831e−03	6.886e−01
UBE2I	−5.019001	3.116453e−06	1.831e−03	6.886e−01
CEMIP	−4.964176	4.139594e−06	1.955e−03	6.886e−01
SNAP25	−4.929729	4.940626e−06	1.987e−03	6.886e−01

GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Steroid Metabolic Process (GO:0008202)	0.15944633	56	3.767e−05	2.022e−01	HSD17B2:14 HSD11B1:107 CYP11A1:201 CYP51A1:375 HSD17B3:409 CYP21A2:412
'De Novo' AMP Biosynthetic Process (GO:0	−0.15293445	4	2.895e−01	9.580e−01	PFAS:128 PAICS:5819 ATIC:5819 ADSS1:5819 NA NA
'De Novo' Post−Translational Protein Fol	0.09781599	15	1.898e−01	9.580e−01	SDF2L1:887 DNAJB13:1097 HSPA14:1742 GAK:1896 ERO1A:2443 ENTPD5:6847
2−Oxoglutarate Metabolic Process (GO:000	0.15259031	13	5.685e−02	9.580e−01	GOT2:166 TAT:169 OGDHL:185 GOT1:484 L2HGDH:1540 IDH1:6847
3'−UTR−mediated mRNA Destabilization (GO	−0.02404704	11	7.825e−01	9.580e−01	DND1:1427 TRIM71:5819 RC3H1:5819 ZFP36L1:5819 ZFP36L2:5819 MOV10:5819
3'−UTR−mediated mRNA Stabilization (GO:0	0.07242310	11	4.057e−01	9.580e−01	MAPK14:1303 HNRNPC:1409 HNRNPA0:1754 DAZL:2202 ZFP36:6847 TARDBP:6847
3'−Phosphoadenosine 5'−Phosphosulfate Me	0.24757905	8	1.532e−02	9.580e−01	SULT1A2:97 SULT1B1:964 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 PAPSS2:6847
5S Class rRNA Transcription By RNA Polym	−0.04055290	5	7.535e−01	9.580e−01	GTF3C2:5819 GTF3C3:5819 GTF3C4:5819 GTF3C5:5819 GTF3C6:5819 NA
7−Methylguanosine RNA Capping (GO:000945	0.20867202	2	3.067e−01	9.580e−01	RNMT:569 CMTR2:6847 NA NA NA NA
7−Methylguanosine mRNA Capping (GO:00063	0.20867202	2	3.067e−01	9.580e−01	RNMT:569 CMTR2:6847 NA NA NA NA
ADP Transport (GO:0015866)	−0.02679134	5	8.356e−01	9.580e−01	SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 SLC25A41:12017 NA
AMP Biosynthetic Process (GO:0006167)	−0.18086332	6	1.250e−01	9.580e−01	PFAS:128 APRT:855 PAICS:5819 ATIC:5819 ADSS1:5819 NUDT2:5819
ATF6−mediated Unfolded Protein Response	0.05763797	5	6.554e−01	9.580e−01	CREBZF:656 MBTPS1:6847 XBP1:6847 ATF6B:6847 DDIT3:6847 NA
ATP Biosynthetic Process (GO:0006754)	0.02728418	13	7.335e−01	9.580e−01	TGFB1:780 ATP5F1C:1803 NUDT2:6847 VPS9D1:6847 ATP5PF:6847 ATP5PD:6847
ATP Metabolic Process (GO:0046034)	0.02856910	25	6.212e−01	9.580e−01	TGFB1:780 ENPP3:1053 BAD:1081 MYH6:1785 ATP5F1C:1803 PARG:2089
ATP Synthesis Coupled Electron Transport	0.11562172	3	4.879e−01	9.580e−01	NDUFB6:949 NDUFS2:6847 NDUFV3:6847 NA NA NA
ATP Transport (GO:0015867)	−0.08235327	9	3.923e−01	9.580e−01	CALHM1:191 SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 CALHM3:5819
AV Node Cell To Bundle Of His Cell Commu	0.04574441	4	7.514e−01	9.580e−01	GJA5:2469 SCN5A:6847 CXADR:6847 GJC3:6847 NA NA
Arp2/3 Complex−Mediated Actin Nucleation	−0.09664430	10	2.900e−01	9.580e−01	WHAMM:727 ARPC5L:796 WASHC1:1166 ARPC1A:1319 ARPC5:5819 ARPC3:5819
B Cell Activation (GO:0042113)	0.06407234	61	8.398e−02	9.580e−01	CD86:34 HDAC9:214 IFNK:316 IL4:679 CASP8:1036 CR2:1071
B Cell Chemotaxis (GO:0035754)	0.05649964	5	6.617e−01	9.580e−01	CYP7B1:734 PIK3CD:6847 HSD3B7:6847 CXCL13:6847 GAS6:6847 NA
B Cell Differentiation (GO:0030183)	0.05405264	44	2.153e−01	9.580e−01	HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAM1:1120 PTPN2:1201
B Cell Homeostasis (GO:0001782)	0.03584862	7	7.426e−01	9.580e−01	TNFRSF13B:400 MEF2C:1121 GAPT:6847 DOCK10:6847 NCKAP1L:6847 TNFAIP3:6847
B Cell Proliferation (GO:0042100)	0.07943161	15	2.870e−01	9.580e−01	IFNK:316 CR2:1071 MEF2C:1121 PTPRC:2183 IL10:6847 CD70:6847
B Cell Receptor Signaling Pathway (GO:00	0.07446488	26	1.890e−01	9.580e−01	SH2B2:544 CTLA4:865 CD38:1042 MEF2C:1121 RFTN1:2025 PTPRC:2183
BMP Signaling Pathway (GO:0030509)	−0.04136061	42	3.542e−01	9.580e−01	LEFTY2:341 LEFTY1:489 GDF10:1012 FAM83G:1081 INHBE:1119 SLC39A5:1205
C−terminal Protein Amino Acid Modificati	−0.08813923	8	3.880e−01	9.580e−01	AGBL4:781 GPLD1:967 AGBL5:5819 AGTPBP1:5819 FOLH1:5819 ATG16L1:5819
C−terminal Protein Deglutamylation (GO:0	−0.13997713	4	3.323e−01	9.580e−01	AGBL4:781 AGBL5:5819 AGTPBP1:5819 FOLH1:5819 NA NA
C−terminal Protein Lipidation (GO:000650	−0.03478644	3	8.347e−01	9.580e−01	GPLD1:967 ATG16L1:5819 ATG7:10875 NA NA NA
C4−dicarboxylate Transport (GO:0015740)	0.11620399	10	2.033e−01	9.580e−01	SLC1A1:216 SLC25A12:618 SLC25A18:992 SLC13A2:1451 SLC25A13:6847 SLC13A5:6847
CD4−positive, Alpha−Beta T Cell Activati	−0.13020632	5	3.133e−01	9.580e−01	NKG7:147 STOML2:5819 TNFSF4:5819 RSAD2:5819 HMGB1:5819 NA
CD40 Signaling Pathway (GO:0023035)	0.03632786	7	7.393e−01	9.580e−01	CD86:34 CD40LG:6847 ITGB1:6847 PHB2:6847 TNIP2:6847 ITGA5:6847
CENP−A Containing Chromatin Assembly (GO	0.04334790	5	7.371e−01	9.580e−01	NASP:1553 MIS18A:6847 CENPI:6847 HJURP:6847 OIP5:6847 NA
COPI Coating Of Golgi Vesicle (GO:004820	−0.05684820	2	7.807e−01	9.580e−01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COPI−coated Vesicle Budding (GO:0035964)	−0.05684820	2	7.807e−01	9.580e−01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COPII Vesicle Coating (GO:0048208)	−0.03438454	14	6.561e−01	9.580e−01	KLHL12:475 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2L:5819 TRAPPC1:5819 TRAPPC2:5819
COPII−coated Vesicle Budding (GO:0090114	−0.01150596	29	8.303e−01	9.580e−01	TBC1D20:311 KLHL12:475 SH2D2A:491 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2L:5819
CRD−mediated mRNA Stabilization (GO:0070	−0.19756073	5	1.261e−01	9.580e−01	IGF2BP1:496 YBX1:1195 DHX9:5819 PABPC1:5819 HNRNPD:5819 NA
Cdc42 Protein Signal Transduction (GO:00	−0.04055702	6	7.308e−01	9.580e−01	SHTN1:5819 WAS:5819 RHOV:5819 NTN1:5819 RHOBTB1:5819 RHOBTB2:5819
DNA Alkylation (GO:0006305)	0.09262387	15	2.144e−01	9.580e−01	DNMT3B:31 GATAD2A:146 BAZ2A:343 HELLS:1649 KMT2E:6847 MTRR:6847

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	−0.29109711	27	1.661e−07	1.075e−03	ADAM15:45 PBXIP1:80 MTX1:85 RUSC1:89 RORC:177 FDPS:188
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	−0.14502570	100	5.637e−07	1.825e−03	COL20A1:30 HELZ2:88 MTG2:97 CDH26:104 FAM217B:130 LAMA5:149
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.13697150	92	5.790e−06	1.250e−02	SULT1A2:97 AKR1D1:117 MGST1:161 CYP2F1:176 COMT:179 CYP11A1:201
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.23916671	26	2.444e−05	3.957e−02	HSD17B2:14 HSD11B1:107 AKR1D1:117 COMT:179 CYP11A1:201 HSD17B3:409
REACTOME_BIOLOGICAL_OXIDATIONS	0.09916207	123	1.503e−04	1.947e−01	ALDH1A1:73 SULT1A2:97 HPGDS:136 MGST1:161 CYP2F1:176 COMT:179
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.19383617	28	3.868e−04	2.784e−01	HSD17B2:14 HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 SERPINA6:585
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.10671382	95	3.319e−04	2.784e−01	SLC36A1:71 SLC36A2:135 SLC12A3:193 SLC1A1:216 SLC6A6:251 SLC7A6:272
REACTOME_METABOLISM_OF_LIPIDS	0.04463701	576	2.908e−04	2.784e−01	HSD17B2:14 SLCO1B1:82 PTGR1:96 HSD11B1:107 AKR1D1:117 HPGDS:136
CAIRO_HEPATOBLASTOMA_DN	0.07440518	196	3.441e−04	2.784e−01	TMEM131L:35 AKR1D1:117 SLC46A3:125 CA2:152 SLC27A5:177 OGDHL:185
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.07628784	180	4.324e−04	2.800e−01	IFNAR1:58 IL1R2:66 TNF:87 TNFSF10:103 TNFSF15:112 IL1RAP:304
NIKOLSKY_BREAST_CANCER_8Q23_Q24_AMPLICON	−0.09228699	111	7.998e−04	3.176e−01	PLEC:24 FAM83H:202 GPR20:254 TRAPPC9:286 OPLAH:293 SCRIB:348
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.43146604	5	8.338e−04	3.176e−01	CA2:152 CA4:253 AQP1:762 SLC4A1:1060 RHAG:2146 NA
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.06778672	219	5.733e−04	3.176e−01	SLC22A16:3 SLC36A1:71 SLCO1B1:82 SLC10A6:83 SLC36A2:135 SLC12A3:193
WP_2Q112_COPY_NUMBER_VARIATION_SYNDROME	0.17941272	30	6.742e−04	3.176e−01	CNNM4:163 ARID5A:424 CNNM3:508 KANSL3:580 COL2A1:605 DUSP2:677
WP_STEROID_HORMONE_PRECURSOR_BIOSYNTHESI	0.36775507	7	7.532e−04	3.176e−01	AKR1D1:117 CYP11A1:201 CYP21A2:412 CYP17A1:883 HSD3B2:1180 SRD5A2:2125
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20027634	24	6.854e−04	3.176e−01	GOT2:166 AGXT:186 CAD:245 IL4I1:373 GLS2:427 GOT1:484
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.21228245	21	7.602e−04	3.176e−01	DNMT3B:31 DNMT1:113 GOT2:166 TAT:169 IL4I1:373 LDHC:408
NIKOLSKY_BREAST_CANCER_17Q21_Q25_AMPLICO	−0.06055546	248	1.075e−03	3.783e−01	FTSJ3:58 HOXB13:67 LRRC59:71 ITGB4:100 ENGASE:119 EVPL:121
WP_CLASSICAL_PATHWAY_OF_STEROIDOGENESIS_	0.27189643	12	1.110e−03	3.783e−01	HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 CYP17A1:883 HSD3B2:1180
BENPORATH_ES_WITH_H3K27ME3	−0.03573796	730	1.177e−03	3.810e−01	CEMIP:10 BNC1:14 PLEC:24 SUSD4:34 ADAM15:45 CAMK2B:46
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	−0.14706730	40	1.297e−03	3.998e−01	PIEZO1:56 CPNE7:138 ZC3H18:408 ANKRD11:431 SPATA2L:545 MVD:604
BENPORATH_SUZ12_TARGETS	−0.03630485	681	1.422e−03	4.187e−01	CEMIP:10 BNC1:14 PLEC:24 SUSD4:34 ADAMTSL3:50 HOXB13:67
REACTOME_GLUCOCORTICOID_BIOSYNTHESIS	0.36870547	6	1.762e−03	4.226e−01	HSD11B1:107 CYP21A2:412 SERPINA6:585 CYP17A1:883 HSD3B2:1180 HSD11B2:6847
WP_ALLOGRAFT_REJECTION	0.11271893	65	1.691e−03	4.226e−01	CD86:34 TNF:87 HLA−DOB:137 FOXP3:324 ABCB1:397 IL1A:608
WP_PROXIMAL_TUBULE_TRANSPORT	0.13658419	44	1.732e−03	4.226e−01	ATP6V0A4:43 SLC36A2:135 CA2:152 SLC1A1:216 CA4:253 ABCB1:397
KEGG_SULFUR_METABOLISM	0.30192728	9	1.710e−03	4.226e−01	SULT1A2:97 SULT1A4:370 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 CHST13:2118
KEGG_TYPE_I_DIABETES_MELLITUS	0.18085786	25	1.753e−03	4.226e−01	CD86:34 TNF:87 HLA−DOB:137 IL1A:608 CD80:787 IFNG:1016
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	−0.04988831	333	1.866e−03	4.316e−01	CHRNB2:26 SHE:40 CAMK2B:46 HTR1D:55 KCNH2:81 YBX2:87
REACTOME_ARACHIDONIC_ACID_METABOLISM	0.14099083	40	2.044e−03	4.564e−01	PTGR1:96 HPGDS:136 GGT1:470 GPX4:500 CYP1A1:528 GGT5:642
REACTOME_CYTOCHROME_P450_ARRANGED_BY_SUB	0.14731339	35	2.573e−03	5.375e−01	CYP2F1:176 CYP11A1:201 CYP51A1:375 CYP21A2:412 CYP27A1:499 CYP1A1:528
KEGG_GRAFT_VERSUS_HOST_DISEASE	0.21133524	17	2.559e−03	5.375e−01	CD86:34 TNF:87 HLA−DOB:137 IL1A:608 CD80:787 IFNG:1016
PID_NFAT_TFPATHWAY	0.14661319	35	2.697e−03	5.458e−01	TNF:87 FOXP3:324 DGKA:353 EGR4:646 IL4:679 CTLA4:865
REACTOME_MEIOTIC_RECOMBINATION	−0.15454536	31	2.912e−03	5.714e−01	MRE11:406 ATM:434 H2AZ2:626 MLH1:818 H3−4:831 H2BC9:1058
REACTOME_TNFS_BIND_THEIR_PHYSIOLOGICAL_R	0.17720343	23	3.271e−03	6.186e−01	TNFSF15:112 TNFRSF13B:400 TNFSF14:613 TNFSF8:708 TNFRSF9:836 FASLG:1216
PYEON_HPV_POSITIVE_TUMORS_UP	0.09915087	73	3.436e−03	6.186e−01	TASOR:41 TAF7L:154 UHRF1:155 TRIM56:411 DTL:423 GLS2:427
KEGG_ALLOGRAFT_REJECTION	0.18445481	21	3.439e−03	6.186e−01	CD86:34 TNF:87 HLA−DOB:137 IL4:679 CD80:787 IFNG:1016
LASTOWSKA_NEUROBLASTOMA_COPY_NUMBER_UP	−0.07130891	138	3.908e−03	6.214e−01	MINK1:51 FTSJ3:58 LRRC59:71 NEURL4:200 CHRNB1:265 P4HB:270
BIOCARTA_HBX_PATHWAY	−0.36844491	5	4.327e−03	6.214e−01	SHC1:353 CREB1:565 LAMTOR5:675 SOS1:955 PTK2B:5819 NA
IVANOVA_HEMATOPOIESIS_MATURE_CELL	0.05934998	203	3.673e−03	6.214e−01	ST3GAL5:84 IL33:127 GCNT1:129 TSPAN8:167 TM4SF4:195 COL1A1:209
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	−0.16105691	26	4.487e−03	6.214e−01	TP53BP2:38 ATM:434 CASP2:465 AIFM2:528 TP73:610 PERP:828

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Dermatitis	0.10566782	138	1.950e−05	6.349e−02	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Leprosy	0.13214844	88	1.908e−05	6.349e−02	TNF:87 TNFSF15:112 HLA−DOB:137 FOXP3:324 IL18RAP:416 CD163:524
Malignant neoplasm of colon stage IV	0.23365136	29	1.347e−05	6.349e−02	PROM1:22 TNFSF10:103 TPX2:286 SMPD1:606 FBLN1:633 VEGFA:756
Hepatitis C	0.05525428	498	3.112e−05	7.599e−02	AFP:1 CLU:17 GGTLC3:29 STAT2:38 IFNAR1:58 IL1R2:66
Severe Dengue	0.19085259	38	4.749e−05	9.275e−02	RBFOX1:36 TNF:87 TLR7:238 VEGFA:756 CTLA4:865 CXCL8:1051
Cystic Fibrosis	0.06059623	373	6.930e−05	9.668e−02	AFP:1 BPIFB1:4 DNMT3B:31 CELA3B:42 TRPV5:55 TNF:87
Systemic Scleroderma	0.06286578	347	6.705e−05	9.668e−02	PRSS55:2 IFNAR1:58 ENG:64 TNF:87 TNFSF10:103 DNMT1:113
Psoriasis	0.05100856	522	8.434e−05	1.030e−01	IL1RL2:8 STAT2:38 TNF:87 TNFSF10:103 ITGAX:104 TNFSF15:112
Balkan Nephropathy	0.35334846	10	1.094e−04	1.188e−01	TNF:87 GSTK1:235 ABCB1:397 ALB:530 SLCO6A1:599 IL1A:608
Goldenhar Syndrome	−0.36617357	9	1.427e−04	1.393e−01	TSC1:59 FRAS1:279 TBC1D20:311 FGF8:316 GRIP1:538 CHD7:768
Hepatitis	0.07592066	201	2.231e−04	1.676e−01	AFP:1 PROM1:22 SLCO1B1:82 TNF:87 TNFSF10:103 AKR1D1:117
IGA Glomerulonephritis	0.08276435	169	2.192e−04	1.676e−01	CD86:34 TNF:87 ITGAX:104 MAN2B1:106 HORMAD2:118 FCGRT:149
Inflammatory dermatosis	0.10099228	115	1.913e−04	1.676e−01	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Renal salt wasting	0.28288815	14	2.484e−04	1.733e−01	CYP11A1:201 CYP21A2:412 SCNN1A:465 CLCNKA:664 SCNN1G:1157 HSD3B2:1180
Kidney Failure, Chronic	0.05924546	310	3.741e−04	1.857e−01	TRPV1:13 CLU:17 CHDH:77 TNF:87 HSD11B1:107 MSR1:115
Acquired aplastic anemia	0.17905557	34	3.056e−04	1.857e−01	TNF:87 HPGDS:136 GSTK1:235 FOXP3:324 HNF1A:559 SLCO6A1:599
Anemia	0.05613830	344	3.926e−04	1.857e−01	DNMT3B:31 ATP2B4:33 IFNAR1:58 ENG:64 TNF:87 TNFSF10:103
Helminthiasis	0.23432224	20	2.875e−04	1.857e−01	PRSS55:2 ITGAX:104 IL33:127 AGXT:186 FOXP3:324 IL4:679
Nonalcoholic Steatohepatitis	0.08576194	146	3.657e−04	1.857e−01	DNMT3B:31 TNF:87 TNFSF10:103 HSD11B1:107 DNMT1:113 IL33:127
Polymorphous light eruption	0.39168099	7	3.327e−04	1.857e−01	TNF:87 HPGDS:136 GSTK1:235 SLCO6A1:599 IL1A:608 ICAM1:1079
Rare Diseases	0.26410762	15	3.992e−04	1.857e−01	DNMT3B:31 AGXT:186 FGFR1:480 TTC7A:837 CPS1:962 FGF23:1181
Carcinoma, Transitional Cell	0.06323810	260	4.892e−04	1.978e−01	DNMT3B:31 ENG:64 TNF:87 TNFSF10:103 DNMT1:113 HPGDS:136
Brucellosis	0.19170889	28	4.497e−04	1.978e−01	RSL1D1:51 TNF:87 PDE4A:130 MEFV:362 IL1A:608 IL4:679
Chronic Kidney Diseases	0.06288187	263	4.897e−04	1.978e−01	CHDH:77 TNF:87 HPGDS:136 AGXT:186 GSTK1:235 NPHS1:242
Dengue Fever	0.10010563	101	5.253e−04	1.978e−01	IFNAR1:58 TNF:87 TNFSF10:103 ITGAX:104 NCL:210 DAXX:315
Hypocalciuria	0.37841226	7	5.266e−04	1.978e−01	TRPV5:55 SLC12A3:193 HNF1A:559 CASR:720 PTH:899 CLCNKB:1477
Encephalomyelitis	0.06318748	254	5.723e−04	2.054e−01	TRPV1:13 CLU:17 CD86:34 TNF:87 TNFSF10:103 HPGDS:136
Septicemia	0.05618789	322	5.889e−04	2.054e−01	CD86:34 TRPV5:55 RELB:62 TNF:87 IL33:127 PLAUR:164
Alcoholic Liver Diseases	0.13097573	57	6.363e−04	2.094e−01	TRPV1:13 NCAN:15 ALDH1A1:73 TNF:87 HPGDS:136 GSTK1:235
Hepatocarcinogenesis	0.05546298	326	6.433e−04	2.094e−01	AFP:1 CLU:17 PROM1:22 DNMT3B:31 TNF:87 PTGR1:96
Congenital dyserythropoietic anemia	0.28201144	12	7.197e−04	2.117e−01	MAN2A1:108 SEC23B:225 LPIN2:442 GATA1:482 AQP1:762 KLF1:990
Hyponatremia	0.17336229	32	6.947e−04	2.117e−01	COMT:179 CYP11A1:201 ABCB1:397 SCNN1A:465 CLCNKA:664 SLC26A3:665
Sepsis	0.05209496	368	6.779e−04	2.117e−01	CD86:34 ATP6V0A4:43 TRPV5:55 RELB:62 TNF:87 IL33:127
Steroid−sensitive nephrotic syndrome	0.32496165	9	7.370e−04	2.117e−01	TNF:87 PLA2G7:654 VEGFA:756 ACE:981 CXCL8:1051 IL12B:1171
Dermatitis, Atopic	0.05535255	316	7.939e−04	2.215e−01	RELB:62 ALDH1A1:73 TNF:87 DNMT1:113 IL33:127 HPGDS:136
Immune thrombocytopenic purpura	0.08466606	130	8.925e−04	2.421e−01	DNMT3B:31 CD86:34 TNF:87 TNFSF10:103 IL33:127 FCGRT:149
Acanthocytosis	0.29915179	10	1.056e−03	2.423e−01	XK:45 VPS13A:206 CAD:245 GATA1:482 KEL:709 PANK2:969
Bacterial Infections	0.07199530	178	9.740e−04	2.423e−01	TNF:87 SLC27A5:177 MPEG1:188 CPD:223 NFE2L2:305 GDE1:356
Diarrhea	0.06205069	237	1.075e−03	2.423e−01	DNMT3B:31 SCN11A:37 TNF:87 AKR1D1:117 SLC12A3:193 CPT1A:293
Female hypogonadism syndrome	−0.20257727	22	1.010e−03	2.423e−01	FGF8:316 ANOS1:321 ATM:434 PROKR2:436 WDR11:615 GNRHR:620

KeggReactome Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_METABOLISM_OF_LIPIDS	0.06841396	650	5.966e−09	1.076e−05	ALB:40 CYP17A1:47 HSD11B1:49 FABP1:59 SPTLC3:67 ELOVL7:97
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.05511916	623	4.302e−06	3.398e−03	TRPV1:4 ANO5:17 ALB:40 SLC22A16:52 SLC38A4:57 SLC13A2:69
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.08431892	232	1.131e−05	3.398e−03	SLC22A16:52 SLC38A4:57 SLC13A2:69 SLC6A6:70 SLC26A3:82 SLC5A12:92
REACTOME_METABOLISM_OF_AMINO_ACIDS_AND_D	0.07528592	299	9.284e−06	3.398e−03	RPL23:5 RPS14:16 BCKDK:22 RPS15A:31 GLS2:36 SDS:37
REACTOME_MEIOTIC_RECOMBINATION	−0.18983971	47	6.914e−06	3.398e−03	H4C15:41 H4C14:67 H4C3:144 H4C2:218 H4C13:273 H2BC15:313
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.24043031	28	1.083e−05	3.398e−03	CYP17A1:47 HSD11B1:49 COMT:65 HSD17B2:124 CYP1A1:159 HSD17B3:195
REACTOME_SPERM_MOTILITY_AND_TAXES	0.42588855	8	3.036e−05	5.598e−03	KCNU1:286 CATSPERD:403 CATSPERB:536 CATSPERG:615 CATSPER2:1018 CATSPER4:1064
REACTOME_BIOLOGICAL_OXIDATIONS	0.10075978	145	3.066e−05	5.598e−03	SULT1A2:14 SULT1A3:24 SULT1A4:28 MGST3:51 COMT:65 CYP2F1:75
REACTOME_SIGNALING_BY_GPCR	−0.05076419	595	3.415e−05	5.598e−03	CAMK2B:18 CDC42:49 TACR2:64 FSHR:125 SSTR4:127 RHOA:128
REACTOME_G_ALPHA_12_13_SIGNALLING_EVENTS	−0.15513436	62	2.481e−05	5.598e−03	RHOA:128 TIAM2:139 ARHGEF11:145 ARHGEF2:173 OBSCN:396 FGD1:437
REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC	0.21880596	30	3.408e−05	5.598e−03	CYP17A1:47 CYP21A2:380 CYP11A1:419 GGT1:457 CYP19A1:597 ACY1:708
REACTOME_MEIOSIS	−0.13678867	75	4.386e−05	5.648e−03	UBE2I:7 H4C15:41 H4C14:67 H4C3:144 SYNE1:167 H4C2:218
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.11833287	101	4.203e−05	5.648e−03	SLC38A4:57 SLC6A6:70 SLC26A3:82 SLC5A12:92 SLC36A1:111 SLC12A1:112
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	−0.15106526	62	4.027e−05	5.648e−03	RPS27A:48 UBA52:88 POLD2:188 EME1:223 RTEL1:277 POLE:341
REACTOME_GENE_SILENCING_BY_RNA	−0.12855410	82	5.975e−05	7.182e−03	POLR2J:32 H4C15:41 RAN:56 H4C14:67 H4C3:144 NUP88:177
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20492863	30	1.038e−04	1.170e−02	GLS2:36 GOT2:55 AGXT:103 CAD:354 CPS1:590 GOT1:704
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.45575982	6	1.108e−04	1.175e−02	CA2:106 AQP1:223 CA4:335 SLC4A1:392 RHAG:1001 CA1:1527
REACTOME_NRAGE_SIGNALS_DEATH_THROUGH_JNK	−0.15134762	54	1.226e−04	1.185e−02	TIAM2:139 ARHGEF11:145 ARHGEF2:173 OBSCN:396 FGD1:437 ARHGEF12:512
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	−0.09675813	133	1.249e−04	1.185e−02	UBE2I:7 H4C15:41 RPS27A:48 H4C14:67 UBA52:88 H4C3:144
REACTOME_HOMOLOGY_DIRECTED_REPAIR	−0.10712008	107	1.361e−04	1.227e−02	UBE2I:7 H4C15:41 RPS27A:48 H4C14:67 UBA52:88 H4C3:144
REACTOME_P75_NTR_RECEPTOR_MEDIATED_SIGNA	−0.11630786	85	2.185e−04	1.876e−02	RPS27A:48 UBA52:88 RHOA:128 TIAM2:139 ARHGEF11:145 ARHGEF2:173
REACTOME_CELL_DEATH_SIGNALLING_VIA_NRAGE	−0.13048924	66	2.537e−04	2.079e−02	RPS27A:48 UBA52:88 TIAM2:139 ARHGEF11:145 ARHGEF2:173 CASP2:326
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.18882716	31	2.775e−04	2.097e−02	SLC38A4:57 SLC6A6:70 SLC36A1:111 SLC7A9:115 SLC6A19:193 SLC6A12:481
REACTOME_SIRT1_NEGATIVELY_REGULATES_RRNA	−0.18870388	31	2.801e−04	2.097e−02	H4C15:41 H4C14:67 H4C3:144 H4C2:218 H4C13:273 H2BC15:313
REACTOME_SENSORY_PROCESSING_OF_SOUND	−0.12733561	68	2.908e−04	2.097e−02	SNAP25:11 PCDH15:103 RIPOR2:192 CTBP2:252 ACTG1:367 CACNB2:368
KEGG_CYSTEINE_AND_METHIONINE_METABOLISM	0.18908808	30	3.417e−04	2.369e−02	SDS:37 TAT:54 GOT2:55 DNMT3B:64 DNMT1:265 CTH:442
KEGG_GLYCINE_SERINE_AND_THREONINE_METABO	0.18539866	31	3.578e−04	2.389e−02	SDS:37 AGXT:103 GNMT:121 CHDH:299 DAO:383 CTH:442
KEGG_OXIDATIVE_PHOSPHORYLATION	0.10767390	92	3.723e−04	2.398e−02	NDUFS5:78 NDUFB5:486 ATP6V0A4:516 NDUFA9:602 COX5B:633 NDUFB8:636
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.18630625	30	4.175e−04	2.509e−02	CYP17A1:47 HSD11B1:49 HSD17B2:124 HSD17B3:195 AKR1B1:369 CYP21A2:380
REACTOME_DEATH_RECEPTOR_SIGNALING	−0.08782126	137	4.106e−04	2.509e−02	UBE2D1:5 RACK1:23 RPS27A:48 BIRC3:75 UBA52:88 RHOA:128
REACTOME_O_GLYCOSYLATION_OF_TSR_DOMAIN_C	−0.16509465	38	4.352e−04	2.531e−02	ADAMTSL3:40 ADAMTS7:62 ADAMTS12:478 THSD4:588 ADAMTSL1:776 THSD1:851
REACTOME_AMYLOID_FIBER_FORMATION	−0.13278525	58	4.803e−04	2.706e−02	H4C15:41 RPS27A:48 H4C14:67 UBA52:88 H4C3:144 H4C2:218
REACTOME_ACTIVATED_PKN1_STIMULATES_TRANS	−0.19373075	27	4.983e−04	2.722e−02	H4C15:41 H4C14:67 H4C3:144 H4C2:218 H4C13:273 H2BC15:313
REACTOME_FATTY_ACID_METABOLISM	0.08151227	154	5.137e−04	2.724e−02	ELOVL7:97 CYP1A1:159 HSD17B3:195 CPT1A:200 PTGS2:268 GGT5:293
REACTOME_NONHOMOLOGOUS_END_JOINING_NHEJ	−0.15396334	42	5.645e−04	2.908e−02	H4C15:41 H4C14:67 H4C3:144 PAXIP1:198 H4C2:218 XRCC5:263
REACTOME_DEVELOPMENTAL_BIOLOGY	−0.03690718	792	5.936e−04	2.973e−02	MAGOH:20 ETF1:31 POLR2J:32 H4C15:41 ELOC:43 RPS27A:48
REACTOME_GLYCEROPHOSPHOLIPID_BIOSYNTHESI	0.09293836	114	6.372e−04	3.098e−02	PISD:109 SLC44A4:184 MIGA1:376 SLC44A2:417 GPCPD1:434 PLA2G1B:441
REACTOME_RHO_GTPASE_CYCLE	−0.05058453	396	6.529e−04	3.098e−02	IQGAP3:1 ACTC1:29 CDC42:49 COPS2:70 RHOA:128 NHS:134
REACTOME_NEURONAL_SYSTEM	−0.05235158	352	8.545e−04	3.845e−02	SNAP25:11 CAMK2B:18 SLITRK4:87 LIN7C:207 ADCY1:209 KCNV2:232
REACTOME_DISEASES_OF_DNA_REPAIR	−0.13955095	48	8.377e−04	3.845e−02	PMS2:91 BLM:354 TOP3A:662 ATM:952 NEIL1:1112 RAD50:1364

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002136 abnormal kidney physiology	0.06933877	245	2.620e−04	1.247e−01	TRPV1:13 TRPV5:55 RELB:62 MAN2A1:108 AGXT:186 SLC12A3:193
MP0009642 abnormal blood homeostasis	0.03516460	1029	6.594e−04	1.290e−01	AFP:1 TRPV1:13 ANGPTL6:53 IFNAR1:58 RELB:62 LRP4:63
MP0009643 abnormal urine homeostasis	0.06781021	200	1.182e−03	1.290e−01	TRPV5:55 MAN2B1:106 MAN2A1:108 AGXT:186 SLC12A3:193 MGAT2:215
MP0005000 abnormal immune tolerance	0.06424395	220	1.306e−03	1.290e−01	CD86:34 TNF:87 TNFSF10:103 ITGAX:104 MAN2A1:108 TNFSF15:112
MP0008469 abnormal protein level	0.05278539	333	1.355e−03	1.290e−01	CD86:34 IFNAR1:58 RELB:62 TNF:87 IQGAP2:124 HPN:126
MP0001663 abnormal digestive system	0.10588405	70	2.340e−03	1.856e−01	RELB:62 COL19A1:322 FOXP3:324 GLG1:437 FGFR1:480 SLC9A4:496
MP0005381 digestive/alimentary phenotyp	0.08312989	92	6.295e−03	3.916e−01	RELB:62 COL1A1:209 COL19A1:322 FOXP3:324 GLG1:437 MYH11:445
MP0003868 abnormal feces composition	0.15762357	24	7.650e−03	3.916e−01	FOXP3:324 PNLIP:389 ALB:530 APC:538 SLC26A3:665 CYP8B1:809
MP0003656 abnormal erythrocyte physiolo	0.17637808	19	7.883e−03	3.916e−01	MAN2A1:108 NFE2L2:305 AQP9:363 NEK1:982 KLF1:990 SLC4A1:1060
MP0000716 abnormal immune system	0.02944361	807	9.050e−03	3.916e−01	DNMT3B:31 CD86:34 STAT2:38 TRPV5:55 IFNAR1:58 RELB:62
MP0002429 abnormal blood cell	0.02762138	984	8.434e−03	3.916e−01	DNMT3B:31 CD86:34 STAT2:38 TRPV5:55 IFNAR1:58 RELB:62
MP0005636 abnormal mineral homeostasis	0.06910297	118	1.035e−02	4.105e−01	SLC12A3:193 CYP11A1:201 MGAT2:215 SLC30A3:284 NFE2L2:305 GATA1:482
MP0000598 abnormal liver morphology	0.03696799	389	1.603e−02	4.770e−01	DNMT3B:31 RELB:62 TNF:87 MAN2B1:106 MAN2A1:108 IQGAP2:124
MP0001765 abnormal ion homeostasis	0.07375443	92	1.536e−02	4.770e−01	TRPV5:55 SLC12A3:193 CYP11A1:201 MGAT2:215 HSPA4L:487 SLC26A3:665
MP0001873 stomach inflammation	0.16293957	19	1.410e−02	4.770e−01	RELB:62 FOXP3:324 SLC9A4:496 IL4:679 CTLA4:865 TOP3B:926
MP0002723 abnormal immune serum	0.03250390	544	1.414e−02	4.770e−01	TRPV1:13 CD86:34 IFNAR1:58 RELB:62 TNF:87 TNFSF10:103
MP0005332 abnormal amino acid	0.06135613	104	3.229e−02	6.404e−01	SLC6A6:251 PEMT:327 GGT1:470 HNF1A:559 ASL:589 BCAT2:696
MP0004134 abnormal chest morphology	0.13416766	22	2.970e−02	6.404e−01	COL1A1:209 APC:538 COL2A1:605 HOXB4:814 FBN1:817 PRKCQ:1078
MP0002139 abnormal hepatobiliary system	0.23403332	7	3.212e−02	6.404e−01	ALDH1A1:73 SLC22A1:903 ABCG8:1002 FABP1:1209 NR1H3:6847 INSIG1:6847
MP0003436 decreased susceptibility to	0.10907843	36	2.399e−02	6.404e−01	TNF:87 IL1R1:456 HIF1A:578 DUSP2:677 TYROBP:765 PTGS1:1183
MP0005195 abnormal posterior eye	−0.04026793	262	2.870e−02	6.404e−01	MOG:18 ADAM15:45 GPR143:99 PRDM16:126 CRIM1:127 XRCC5:161
MP0005391 vision/eye phenotype	−0.05433502	139	2.907e−02	6.404e−01	XRCC5:161 DMD:167 EGFLAM:236 MMP9:266 DPH1:276 RGS9BP:318
MP0002118 abnormal lipid homeostasis	0.03662742	335	2.578e−02	6.404e−01	ANGPTL6:53 ST3GAL5:84 TNF:87 HSD11B1:107 RORA:170 GALC:173
MP0002420 abnormal adaptive immunity	0.02536553	720	3.153e−02	6.404e−01	CD86:34 STAT2:38 IFNAR1:58 MEPE:61 RELB:62 TNF:87
MP0003635 abnormal synaptic transmissio	−0.03590059	314	3.390e−02	6.455e−01	CHRNB2:26 CAMK2B:46 CHRNA3:129 CASK:157 DAGLA:181 AFF2:259
MP0000609 abnormal liver physiology	0.03174742	319	5.882e−02	6.992e−01	TRPV1:13 RELB:62 TNF:87 TNFSF10:103 HSD11B1:107 MAN2A1:108
MP0001529 abnormal vocalization	−0.14857544	14	5.456e−02	6.992e−01	WFS1:510 DHCR7:900 TACR1:1045 CHRM2:1352 DRD2:5819 ACHE:5819
MP0001845 abnormal inflammatory respons	0.02839479	434	5.209e−02	6.992e−01	TRPV1:13 IFNAR1:58 RELB:62 TNF:87 TNFSF10:103 MAN2A1:108
MP0002452 abnormal antigen presenting	0.02652697	489	5.583e−02	6.992e−01	CD86:34 STAT2:38 IFNAR1:58 MEPE:61 RELB:62 TNF:87
MP0010155 abnormal intestine physiology	0.05585952	113	4.241e−02	6.992e−01	RELB:62 IL33:127 SIRT6:260 NFE2L2:305 FOXP3:324 APOA4:449
MP0005197 abnormal uvea morphology	−0.06968440	66	5.169e−02	6.992e−01	CHRNB2:26 ADAM15:45 GPR143:99 CHRNA3:129 PNN:326 DTNBP1:500
MP0008058 abnormal DNA repair	−0.09872517	34	4.703e−02	6.992e−01	PAXIP1:54 PMS2:76 XRCC5:161 TRIP13:368 APEX1:843 DKC1:1141
MP0002135 abnormal kidney morphology	0.03083723	378	4.737e−02	6.992e−01	TRPV1:13 HSD17B2:14 RELB:62 LRP4:63 MAN2B1:106 MAN2A1:108
MP0000639 abnormal adrenal gland	0.07493246	56	5.367e−02	6.992e−01	HSD11B1:107 CYP11A1:201 CHGB:405 APC:538 ACD:584 E2F1:954
MP0001486 abnormal startle reflex	−0.06558057	66	6.708e−02	6.992e−01	PCDH15:43 ABCA2:57 OTOF:158 ATE1:251 NTSR1:524 DVL1:785
MP0005253 abnormal eye physiology	−0.06620717	69	5.883e−02	6.992e−01	XRCC5:161 DMD:167 EGFLAM:236 MMP9:266 RGS9BP:318 AHI1:419
MP0009764 decreased sensitivity to	0.07582377	49	6.757e−02	6.992e−01	TNF:87 IL33:127 BID:441 CYP1A1:528 HIF1A:578 HP:666
MP0001756 abnormal urination	0.06625271	70	5.688e−02	6.992e−01	TRPV1:13 TRPV5:55 MYH11:445 ALB:530 HNF1A:559 CTSA:579
MP0001819 abnormal immune cell	0.02251568	723	5.588e−02	6.992e−01	CD86:34 STAT2:38 IFNAR1:58 MEPE:61 RELB:62 TNF:87
MP0002109 abnormal limb morphology	−0.03591714	275	4.596e−02	6.992e−01	DYNC2H1:6 DISP1:95 ZBTB16:98 CRIM1:127 LAMA5:149 KIF7:156